

SHIXUAN LIN

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Results-driven AI Master's student at National Cheng Kung University (NCKU) specializing in Computer Vision, Reinforcement Learning, and LLM-based applications. Proven track record in developing end-to-end solutions, from video captioning (VolleyCC) to autonomous robotic tracking systems. Proficient in PyTorch, Stable Baselines3, and TensorRT, with a strong focus on bridging the gap between theoretical research and practical deployment. Seeking a summer internship to leverage my expertise in multimodal learning and autonomous agents."

EDUCATION

M.S. in School of Computing (Artificial Intelligence)

2025 – 2027 (Expected)

National Cheng Kung University

Tainan, Taiwan

- **Core Courses:** Computer Vision Practice and Deep Learning A+ | Machine Learning A | Image Recognition and Robotics A | Digital IC Design A-
- Focusing on Computer Vision, Reinforcement Learning, and LLM applications.
- Developing **VolleyCC** (Volleyball Comprehension Captioning) to generate descriptive narratives from game footage.
- Implementing **RECOBOT**, an automatic tissue-catching robot based on JetBot featuring YOLO and Kalman Filter.

BBA in Industrial and Information Management

2021 - 2025

National Cheng Kung University

Tainan, Taiwan

- **Core Courses:** numerical optimization A+ | E-Commerce A+ | Programming and Application A-

EXPERIENCE

NCKU HSU Lab | ComputerVision Developer

Feb 2024 - Now

- Developed and deployed a real-time basketball player tracking system in the cloud, enabling accurate and efficient analysis during games.
- Utilized YOLO for object detection and ByteTrack for multi-object tracking, optimizing performance in high-occlusion environments like basketball courts.
- Collected and labeled 700 images from the Taiwan Professional Basketball League (P.L.G.), achieving 84% mAP, 80% precision, and 77% recall.

Image Website Development | NCKU Freshman Camp(Front-end developer & PM)([link](#))

Feb 2023 - Aug 2023

- Led the development of a website for NCKU Freshman Camp, a student-led initiative aiding freshmen transition to university life.
- Worked as a PM and Front-end developer, Collaborate with other students possessed with back-end development, designer, integrate into comprehensive website with reliable stability which can bear 3000+ daily visits network traffic.
- Achieved exceptional performance metrics,Google Analytic reported 7663 new users and 790K event counts, through effective team coordination and project management.

Engineering Component Recommendation System | NSTC Project (Software Engineer)

Dec 2024 - Oct2 025

- Developed a recommendation system for engineering components under the NSTC project led by Professor Chin-Feng Lai, focusing on high-frequency and low-frequency magnetic component training.
- Designed and implemented an Vector/linear routing algorithm and utilized python scripting for data processing that suggests optimal components based on multiple engineering parameters, improving selection efficiency for researchers and engineers.
- Collaborated with a multidisciplinary team to integrate the system into the training platform, ensuring usability and accessibility for educational and research purposes.

SKILLS

Programming Language	python(Pytorch, Pandas, OpenCV,etc.),HTML/CSS/JS ,C++,Arduino
Database	MySQL,Firestore
Software	Git,GitHub,Linux tools

PROJECTS

GOGORO LLM AI Agent for Electric Vehicle Support | Cloud Rising Taiwan Generative AI

May 2024 - May 2024

Application Competition

- Tech Stack:AWS Segemaker,AWS S3,Python
- Developed an AI Agent based on GOGORO's owner manuals, significantly improving customer support by accurately answering electric vehicle-related queries.
- Leveraged AWS Sagemaker and LangChain frameworks to streamline development and integrate open-source models from Hugging Face, enhancing AI deployment efficiency.
- Applied Retrieval-Augmented Generation (RAG) and Fine-tuning techniques to improve response accuracy for complex vehicle issues.

Yoga at Home – Real-time Video Call and Posture Tracking System

2024

- Tech Stack: Firebase, JavaScript
- Developed a system combining WebRTC and MediaPipe Pose to enable real-time video calls with skeleton tracking, providing users with posture correction to reduce the risk of injury during exercise.
- Integrated WebRTC for low-latency audio and video communication between users, ensuring smooth, real-time interaction.
- Utilized MediaPipe Pose for accurate human pose detection and visual feedback through skeleton tracking displayed on a canvas.

Financial Data Analysis Web Tool | *Personal Project*

Feb 2025

- Tech Stack: SQL, Python, Selenium, Django, SMA, Docker, JavaScript, Highcharts Stock.
- Developed a web-based financial data analysis tool that fetches stock price data from Yahoo Finance and visualizes key financial metrics using Highcharts Stock.
- Implemented automated stock data retrieval with Selenium, handling JavaScript-rendered financial pages and integrating Backtrader for backtesting trading strategies.
- Designed a recommendation system using Simple Moving Averages (SMA) and other quantitative metrics to help users identify trading opportunities.
- Containerized the application with Docker and deployed it via Docker Compose to ensure a consistent and scalable environment.

Competitions

National Finalist, Ming Chiu Innovation Competition (Computing Innovation Category)
Taipei, Taiwan

Multi-Object Basketball
Player Tracking System

4/2024 - 8/2024

- Developed an AI-powered system capable of real-time tracking of basketball players during games, reducing analysis time for players and coaches.
- Utilized **YOLO** (You Only Look Once) for object detection and **ByteTrack** for multi-object tracking, optimizing tracking in high-occlusion environments.
- Collected and labeled 700 images from the Taiwan Professional Basketball League (P.L.G.), achieving **84% mAP, 80% precision, and 77% recall**.
- Technical Skills: Object detection and multi-object tracking, **YOLO**, **ByteTrack**, computer vision, deep learning.

Certificates

Google Cloud Computing and Generative AI (GenAI) Development Certification

Google Digital Talent
Training Program

2024

Languages

English Professional proficiency|TOELC 815